Automotive Manufacturing 🏭

Junior Automotive Engineering Badge — Girls Build Bots

Presented by your robotics mentors!

The scale problem 📈

One great prototype isn’t a car company.
Factories make **thousands** — identical, and safe.

How?

The Assembly Line 🔗

Each worker **masters one step** as the product flows past.

Today the troop becomes a factory — and we'll **measure it** like one.

Round 1: Craft Production 

Build one complete car **alone**. Every step. **8 minutes.**

Then we tally: - How many finished? - Line them up — how **identical** are they?

The honest results 📊

Slow. And every car is a little different.

That’s craft production.

Round 2: The Assembly Line 🏭

One Scout (or pair) per step, plus a **Quality Control inspector** with the spec card.

10 minutes. Go!

QC rejects anything missing a part — it goes back to the station that missed it.

Defects are information 📋

Not shame.

QC rejects **kindly and specifically**: > “Missing left headlight.”

Never “you did it wrong.”

Would you ride in a car nobody inspected?

Read the board 📊

Round	Made	Passed QC	Defects
Solo			
Line			

More cars. More consistent. **And one station drowning in a pile.**

That’s a bottleneck. 🍾

Kaizen Round: improve the line 🔧

Real factories **never run the same line twice**. That's **continuous improvement**.

Team huddle: - Where was the **bottleneck**? - What fixes it? Move a person? Split a step? Pre-stage supplies?

Apply your best ideas. Run 10 more minutes.

Compare all three rounds 🏆

Solo → Line → Improved Line

The improvement round almost always wins big.

And YOU engineered that gain.

Robot moment 

Build season runs like this: - **Parallel stations** — wiring, programming, machining - **Inspections** before competition - Constant **process improvements** between events

Debrief 🎯

1. Why was the line faster?
2. Why did QC matter?
3. What did **your** improvement change?

Count the total fleet — manufacturing is a team sport. 🎉

What you accomplished ✅

- Explained why factories use assembly lines instead of solo builds
- Worked a station and saw how quality control catches defects
- Identified a bottleneck from real production data
- Proposed an improvement, tested it, and compared the rounds

Keep going at home! 

- Run a line at home — cards, sandwiches, anything — and time it
- Find the bottleneck in a routine you do every day
- Ask your troop leader about your next STEM badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.